

Introduction to Logic

Assignment 4

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Problem 1

Table 1 lists some well-known logical equivalences in propositional logic.

Theorem 1 (Replacement Theorem) *Suppose ϕ is a formula and ψ is a subformula of ϕ . And suppose ψ' is a formula such that $\psi \equiv \psi'$. If ϕ' denotes the formula resulted from replacing an occurrence of ψ in ϕ by ψ' , then $\phi \equiv \phi'$.*

The Replacement Theorem allows us to convert a formula into an equivalent one by replacing some subformula ψ in the original formula by any formula equivalent to ψ . The following example shows that the formulas $\neg(\neg p \wedge \neg q)$ and $p \vee q$ are logically equivalent by using repeated applications of the Replacement Theorem and the logical equivalences in Table 1.

Example 1

$$\begin{aligned}\neg(\neg p \wedge \neg q) &\equiv \neg(\neg p) \vee \neg(\neg q) && \text{by E16} \\ &\equiv p \vee \neg(\neg q) && \text{by E9} \\ &\equiv p \vee q && \text{by E9}\end{aligned}$$

By applying the Replacement Theorem and the logical equivalences listed in Table 1, show (as in the previous example) that each pair of formulas below are logically equivalent.

- (a) $\neg(p \rightarrow q)$ and $p \wedge \neg q$
- (b) $(p \wedge q) \vee (\neg p \wedge \neg q)$ and $(\neg p \vee q) \wedge (p \vee \neg q)$
- (c) $p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \rightarrow r)$
- (d) $(q \vee \neg p) \rightarrow (q \wedge p)$ and $p \wedge (r \rightarrow p)$

Table 1: Some Logical Equivalences		
	Equivalences	Name
E1	$\phi \wedge \top \equiv \phi$	Identity Laws
E2	$\phi \vee \perp \equiv \phi$	
E3	$\phi \wedge \perp \equiv \perp$	Domination Laws
E4	$\phi \vee \top \equiv \top$	
E5	$\phi \wedge \neg\phi \equiv \perp$	Complement Laws
E6	$\phi \vee \neg\phi \equiv \top$	
E7	$\phi \wedge \phi \equiv \phi$	Idempotent Laws
E8	$\phi \vee \phi \equiv \phi$	
E9	$\neg(\neg\phi) \equiv \phi$	Double Negation Law
E10	$\phi \wedge \psi \equiv \psi \wedge \phi$	Commutative Laws
E11	$\phi \vee \psi \equiv \psi \vee \phi$	
E12	$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$	Associative Laws
E13	$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$	
E14	$\phi \wedge (\psi \vee \chi) \equiv (\phi \wedge \psi) \vee (\phi \wedge \chi)$	Distributive Laws
E15	$\phi \vee (\psi \wedge \chi) \equiv (\phi \vee \psi) \wedge (\phi \vee \chi)$	
E16	$\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$	De Morgan's Laws
E17	$\neg(\phi \vee \psi) \equiv \neg\phi \wedge \neg\psi$	
E18	$\phi \wedge (\phi \vee \psi) \equiv \phi$	Absorption Laws
E19	$\phi \vee (\phi \wedge \psi) \equiv \phi$	
E20	$\phi \rightarrow \psi \equiv \neg\phi \vee \psi$	
E21	$\phi \leftrightarrow \psi \equiv (\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)$	

(a) $\neg(p \rightarrow q)$ and $p \wedge \neg q$

$$\begin{aligned}
 \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \text{E20} \\
 &\equiv \neg(\neg p) \wedge \neg q && \text{E17} \\
 &\equiv p \wedge \neg q \quad \# && \text{E9}
 \end{aligned}$$

(b) $(p \wedge q) \vee (\neg p \wedge \neg q)$ and $(\neg p \vee q) \wedge (p \vee \neg q)$

$$\begin{aligned}
 &\overset{\phi}{(p \wedge q)} \vee \overset{\psi}{(\neg p \wedge \neg q)} \quad \overset{\chi}{} \\
 &\equiv [(p \wedge q) \vee \neg p] \wedge [(p \wedge q) \vee \neg q] && \text{E15} \\
 &\equiv [\neg p \vee (p \wedge q)] \wedge [\neg q \vee (p \wedge q)] && \text{E11, E11} \\
 &\equiv [(\neg p \vee p) \wedge (\neg p \vee q)] \wedge [(\neg q \vee p) \wedge (\neg q \vee q)] && \text{E15, E15} \\
 &\equiv [(p \vee \neg p) \wedge (\neg p \vee q)] \wedge [(\neg q \vee p) \wedge (q \vee \neg q)] && \text{E11, E11} \\
 &\equiv [\top \wedge (\neg p \vee q)] \wedge [(\neg q \vee p) \wedge \top] && \text{E6, E6} \\
 &\equiv [\top \wedge (\neg p \vee q)] \wedge [\top \wedge (\neg q \vee p)] && \text{E10} \\
 &\equiv (\neg p \vee q) \wedge (\neg q \vee p) && \text{E1, E1} \\
 &\equiv (\neg p \vee q) \wedge (p \vee \neg q) \quad \# && \text{E11}
 \end{aligned}$$

(c) $p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \rightarrow r)$

$$\begin{aligned}
 p \rightarrow (q \rightarrow r) &\equiv p \rightarrow (\neg q \vee r) && \text{E20} \\
 &\equiv \neg p \vee (\neg q \vee r) && \text{E20} \\
 &\equiv (\neg p \vee \neg q) \vee r && \text{E13} \\
 &\equiv (\neg q \vee \neg p) \vee r && \text{E11} \\
 &\equiv \neg q \vee (\neg p \vee r) && \text{E13} \\
 &\equiv q \rightarrow (\neg p \vee r) && \text{E20} \\
 &\equiv q \rightarrow (p \rightarrow r) \quad \# && \text{E20}
 \end{aligned}$$

(d) $(q \vee \neg p) \rightarrow (q \wedge p)$ and $p \wedge (r \rightarrow p)$

$$(q \vee \neg p) \rightarrow (q \wedge p)$$

$$\begin{aligned} &\equiv \neg(q \vee \neg p) \vee (q \wedge p) && \text{E20} \\ &\equiv [\neg q \wedge \neg(\neg p)] \vee (q \wedge p) && \text{E17} \\ &\equiv (\neg q \wedge p) \vee (q \wedge p) && \text{E9} \\ &\equiv (p \wedge \neg q) \vee (p \wedge q) && \text{E10, E10} \\ &\equiv p \wedge (\neg q \vee q) && \text{E14} \\ &\equiv p \wedge T && \text{E11, E6} \\ &\equiv p && \text{E1} \end{aligned}$$

$$p \wedge (r \rightarrow p)$$

$$\begin{aligned} &\equiv p \wedge (\neg r \vee p) && \text{E20} \\ &\equiv p \wedge (p \vee \neg r) && \text{E11} \\ &\equiv p && \text{E18} \end{aligned}$$

$$\therefore (q \vee \neg p) \rightarrow (q \wedge p) \equiv p \wedge (r \rightarrow p)$$